

Finite Element Simulation and Validation of Ultrasonic Wave Propagation for Submerged Coated Metal Composite Structures

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Abstract—In immersion ultrasonic non-destructive testing of metal pipelines with protective coatings, the analysis of echo signals in the time domain is complicated due to the complex multiple reflections and attenuation of ultrasound in the multilayered medium composed of water, the coating, and the metal. To facilitate echo signal analysis, a two-dimensional finite element simulation model is established in COMSOL. An excitation signal is used to generate longitudinal waves in the water-immersed coating-metal composite structure, simulating the spatiotemporal propagation characteristics of ultrasound in the “water-coating-metal” multilayer medium and obtaining pulse-echo ultrasonic signals. The multiple reflection and transmission processes of ultrasound at the interfaces of the multilayered medium are captured, providing a model basis for understanding the time-domain distribution characteristics of the echo signals. Experimental verification demonstrates that the simulated echoes from the multilayer composite are consistent with the pulse-echo characteristics received by the transducer.

Keywords—Ultrasonic measurement, Pulse-echo method, Finite element simulation, Composite Structures

I. INTRODUCTION

Global energy demand has driven the development of subsea pipelines, which suffer internal substrate damage from corrosion and wear in harsh environments^[1-2]. External protective coatings mitigate corrosion but form a physical barrier, impeding the detection of internal substrate damage. Nondestructive testing of coated metal structures is thus crucial for engineering safety.

Traditional detection methods have inherent flaws in this detection task, and ultrasonic testing features high sensitivity and strong penetration^[3-4]. Existing studies have explored ultrasonic applications in pipeline detection^[5-6], yet research on non-contact ultrasonic propagation in coated structures is scarce.

Finite element simulation accurately captures the reflection, transmission and attenuation of ultrasonic waves at water-coating-metal interfaces, a process hard to directly observe via experiments. This study takes steel-based coated composite structures as the research object, adopts ultrasonic liquid-solid coupling method and pulse-echo based non-contact transceiver mode, and uses finite element simulation to clarify ultrasonic propagation mechanisms and analyze the characteristics of echo signals and acoustic fields. Experimental verification is

conducted under consistent simulation conditions, yielding identical echo waveforms, which lays a theoretical foundation for the detection of internal substrate damage in coated metal structures.

II. ULTRASONIC MEASUREMENT PRINCIPLE FOR COMPOSITE STRUCTURES

A. Wave Propagation Model at Medium Interfaces

The propagation of ultrasonic waves requires a medium. If the medium is homogeneous and unbounded, the ultrasonic wave will continue to propagate along the incident direction until its energy is fully dissipated. When the wave encounters an interface between two different media during propagation, its energy is partitioned based on the acoustic impedance of the two media, resulting in the formation of two types of waves: a reflected wave and a transmitted wave.

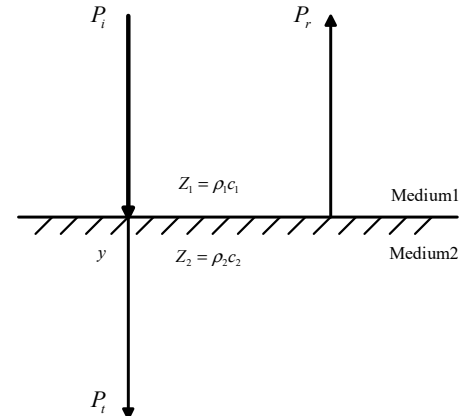


Fig. 1. Schematic diagram of ultrasonic wave propagation at a medium interface

As shown in Fig.1, when an ultrasonic wave is incident from Medium 1 into Medium 2, assuming both Medium 1 and Medium 2 are isotropic and homogeneous materials, the plane wave Equation for the acoustic wave propagating along the vertical y-direction is given by Equation (1):

$$\frac{\partial^2 p}{\partial x^2} = \frac{1}{c_0^2} \frac{\partial^2 p}{\partial t^2} \quad (1)$$

Solving the one-dimensional acoustic wave Equation (1) within the medium yields the acoustic pressure p_1 , as shown in Equation (2):

$$p_1 = p_i + p_r = Ae^{j(\omega t - k_1 y)} + Be^{j(\omega t + k_1 y)} \quad (2)$$

The first term in the above Equation represents the wave propagating in the positive y-direction, i.e., the incident wave p_i , where the constant A is the amplitude of the incident wave. The second term represents the wave propagating in the negative y-direction, which is the reflected wave p_r generated in Medium 1 after the incident wave encounters the medium interface.

The general solution for the acoustic field p_2 in Medium 2 still follows the form of (2). Since Medium 2 extends infinitely, no wave propagating in the negative y-direction exists. Therefore, only the first term in (2) needs to be retained, which corresponds to the transmitted wave p_t , as given by Equation (3):

$$p_2 = p_t = ce^{j(\omega t - k_2 y)} \quad (3)$$

Using Equation (4), the particle velocities in medium 1 and medium 2 can be derived from the sound pressure:

$$v_y = -\frac{1}{\rho_0} \int \frac{\partial p}{\partial y} dt \quad (4)$$

In addition, at the interface $y=0$, the normal particle velocity and the sound pressure are continuous, as shown in Equation (5):

$$\left. \begin{aligned} (p_1)_{y=0} &= (p_2)_{y=0}, \\ (v_1)_{y=0} &= (v_2)_{y=0}. \end{aligned} \right\} \quad (5)$$

Therefore, the particle velocities in medium 1 and medium 2 are given by Equation (6):

$$\left. \begin{aligned} v_1 &= v_i e^{j(\omega t - k_1 y)} + v_r e^{j(\omega t + k_1 y)} \\ v_2 &= v_t e^{j(\omega t - k_2 y)} \end{aligned} \right\} \quad (6)$$

Using Equations (2), (3), and (6), the ratio of the incident sound pressure to the reflected sound pressure at the interface, as well as the ratio of the incident sound pressure to the transmitted sound pressure, can be obtained. Similarly, the proportional relationships for particle velocity can be derived. The specific distribution and variation are determined by the acoustic impedance on both sides of the interface^[7]. The sound pressure reflection coefficient and transmission coefficient can be expressed as Equation (7), and the particle velocity reflection coefficient and transmission coefficient can be expressed as Equation (8).

$$r_p = \frac{p_{ra}}{p_{ia}} = \frac{Z_2 - Z_1}{Z_2 + Z_1}, t_p = \frac{p_{ta}}{p_{ia}} = \frac{2Z_2}{Z_1 + Z_2} \quad (7)$$

$$r_v = \frac{v_{ra}}{v_{ia}} = \frac{Z_1 - Z_2}{Z_1 + Z_2}, t_v = \frac{v_{ta}}{v_{ia}} = \frac{2Z_1}{Z_1 + Z_2} \quad (8)$$

where Z_1 and Z_2 denote the acoustic impedances of the two media on either side of the interface.

B. Reflection Wave Characteristics

This study investigates the changes in acoustic signals associated with the underlying steel in structures coated with an external anti-corrosion layer. Fig.2 shows a schematic diagram of acoustic wave propagation in the coating-steel layered material, where the probe is immersed in water, emitting sound waves vertically downwards and receiving reflected echoes from different medium interfaces.

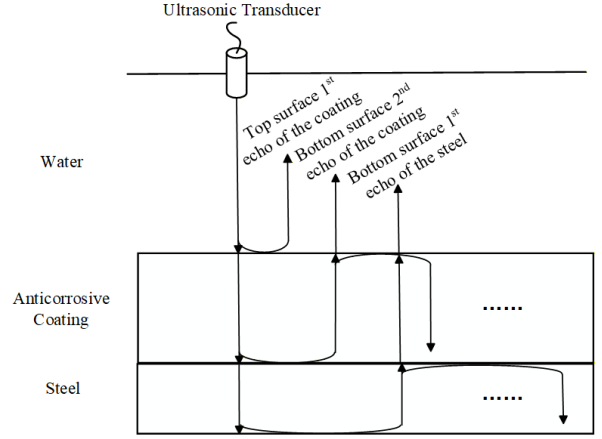


Fig. 2. Schematic diagram of acoustic wave propagation in the coating-steel structure

Based on the acoustic wave propagation schematic in Fig.2 and combined with Equations (7) and (8), the phase changes of the acoustic wave during incidence and reflection at different medium interfaces can be analyzed. For example, when an ultrasonic signal is incident from the anti-corrosion layer (with lower acoustic impedance) into the steel block (with higher acoustic impedance), the interface between medium 1 and medium 2 acts as a hard boundary. Consequently, at this hard boundary, the phase of the reflected wave's particle velocity differs by 180° from that of the incident wave's particle velocity, while the acoustic pressure of the reflected wave is in phase with that of the incident wave. Other interface incidence and reflection scenarios can be analyzed similarly.

III. FINITE ELEMENT SIMULATION

A. Finite Element Model

When ultrasonic waves propagate from water into the workpiece, a two-dimensional model can be employed for the study. This is because the ultrasonic probe is typically circular, and the multilayer workpiece possesses a layered structure with no radial variation in the thickness of each layer along the probe's central axis. Furthermore, ultrasonic waves primarily propagate as longitudinal waves within such layered structures. The reflection and transmission processes occur predominantly along the "probe axis-workpiece depth" direction (with no radial differences), and the physical field distribution is symmetric about the axis. In the model, region A is assigned to the Transient Pressure Acoustics module, regions B and C to the Solid Mechanics module, and region D is a Perfectly Matched Layer (PML).

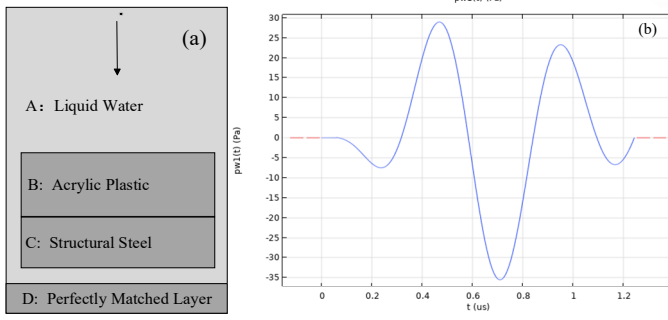


Fig. 3. (a) Finite element simulation geometric model, (b) Excitation waveform at $f=2.25\text{MHz}$

B. Material and Parameter Definitions

In the geometric model shown in Fig.3(a), region A is the liquid water domain with a width of 44 mm and a height of 53 mm; region B is acrylic plastic with a width of 40 mm and a height of 15 mm; region C is structural steel with a width of 40 mm and a height of 10 mm; region D is the PML with a width of 44 mm and a height of 2 mm. The specific physical properties of each material are listed in the table1. And the frequency of the ultrasonic excitation signal is set to 2.25 MHz.

The design of the finite element mesh size directly determines the simulation accuracy and computational efficiency. The core principle of its design is based on the numerical discretization criterion relative to the acoustic wavelength. The mesh must be able to accurately reproduce the propagation, reflection, and transmission characteristics of the acoustic waves, avoiding wave velocity distortion and signal distortion caused by an overly coarse mesh, or a surge in computational load caused by an overly fine mesh^[8]. The selection of mesh size should follow the principle of fully capturing particle vibrations within the wavelength while choosing a size as small as possible, without exceeding the computer's capacity.

C. Ultrasonic Excitation Signal and Boundary Conditions

In practical engineering, an ultrasonic transducer converts an electrical excitation signal into mechanical vibration, which then propagates through a coupling medium (such as water) to form ultrasonic waves^[9]. In the COMSOL simulation, the vibration effect of the transducer is simulated through an equivalent load excitation. The excitation function used here was obtained by sampling the signal from an actual pulse generator connected to an Olympus 2.25 MHz probe using an oscilloscope. The acquired time-domain discrete function was then modeled using

a Fourier regression model. The modeling expression is given as follows:

$$f(x) = a_0 + a_1 \cos(xw) + b_1 \sin(xw) + a_2 \cos(2xw) + b_2 \sin(2xw) + \dots \quad (9)$$

By selecting an appropriate Fourier regression order according to Equation (9), the function can be fitted. Fig.3(b) shows the fitted excitation function, with an R^2 result of 0.977. This excitation function is applied as a plane wave radiation incident from above the water surface at the top of the model in Fig.3(a), achieving an R^2 of 0.977.

To observe the acoustic field variations more clearly and intuitively, the side boundaries of the finite element model in Fig.3(a) are set as plane wave radiation boundaries to simulate propagation in an infinite medium, ensuring no reflected echoes from these sides. A PML is set below the water layer in the model to prevent spurious reflections from the computational domain boundaries from interfering with the echo signals, while also simulating the presence of continuously flowing liquid inside the steel in a real environment.

D. Simulation Results

Extracting the spatial distribution features of acoustic pressure and acceleration from the 2D plot groups allows for intuitive verification of the echo arrival sequence in the time domain. For example, Fig.4 show the two-dimensional distributions of acoustic pressure and acceleration captured at different time instants, corresponding to the four different moments. Based on Equations (7) and (8) and the material properties listed in Table 1, the reflection and transmission coefficients at each interface can be calculated, as summarized in Table 2. These coefficients provide a quantitative basis for understanding the energy partitioning at each interface during the propagation process.

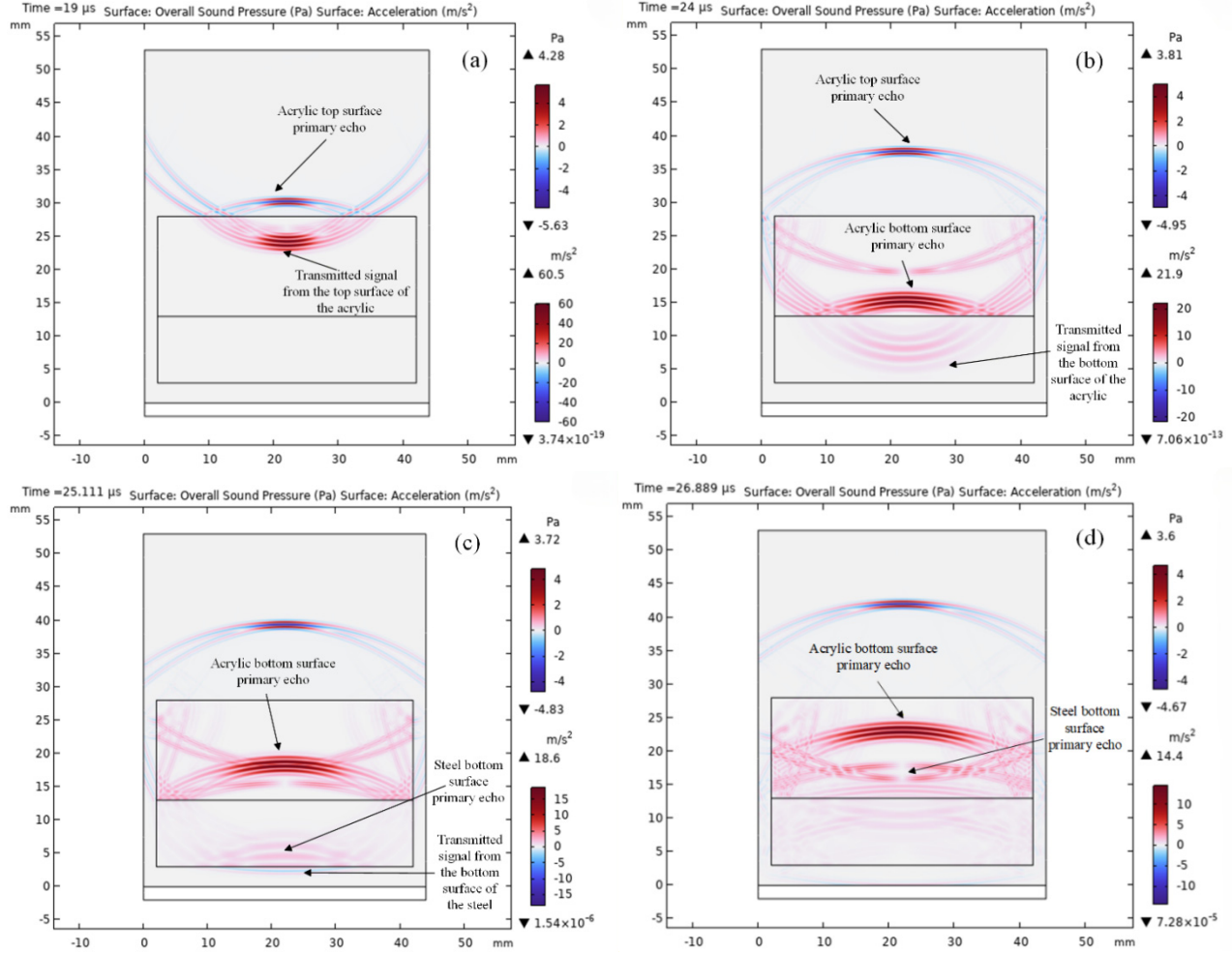
Fig.4(a) presents the acoustic pressure distribution in the water layer and the acceleration distribution in the acrylic layer at $t=19 \mu\text{s}$. At this moment, the excitation signal has propagated through the water domain. At the interface between the water and acrylic layers, partial transmission and reflection of the acoustic wave occur due to the impedance mismatch. As shown in Table 2, the acoustic pressure reflection coefficient at the water–acrylic interface is 0.36 and the transmission coefficient is 1.36. This means that 36% of the incident acoustic pressure is reflected back toward the probe, while the transmitted acoustic pressure is amplified to 1.36 times the incident value. The acoustic impedance of acrylic is approximately 2.15 times that of water.

Table1. Schematic diagram of acoustic wave propagation in the coating-steel structure

Material \ Parameter	Density [kg/m ³]	Pressure Wave Velocity [m/s]	Shear Wave Velocity [m/s]	Acoustic Attenuation [dB/m]
Liquid Water	997.29	1488.63		170.92
Acrylic Plastic	1190	2679.01	1286.99	89.89
Structural Steel	7340	6122.85	3273	447.24

Table2.Reflection and transmission at each interface

Interface \ Parameter	Acoustic Pressure		Particle Velocity	
	Reflection Coefficient	Transmission Coefficient	Reflection Coefficient	Transmission Coefficient
Water-Acrylic	0.36	1.36	-0.36	0.64
Acrylic-Steel	0.86	1.86	-0.86	0.13
Steel-Water	-0.93	0.06	0.93	1.93

Fig. 4. Acoustic pressure and acceleration intensity at different times. (a) $t=19 \mu\text{s}$, (b) $t=24 \mu\text{s}$, (c) $t=25.111 \mu\text{s}$, (d) $t=26.889 \mu\text{s}$

This moderate impedance contrast allows a considerable portion of energy to penetrate into the acrylic layer, which is essential for detecting deeper structures beneath the coating.

Fig.4(b) depicts the further reflection and transmission of the acoustic wave that has entered the acrylic layer at its interface with the steel block. Since the longitudinal wave speed in the structural steel is nearly three times that in the acrylic, the ultrasonic signal propagates much faster within the steel layer. At the acrylic-steel interface, the impedance mismatch is far more severe: the acoustic impedance of steel is roughly 14 times that of acrylic. Consequently, the acoustic pressure reflection coefficient reaches 0.86 and the particle velocity transmission coefficient is only 0.13, indicating that the vast majority of energy is confined within the acrylic layer. This explains why

the echoes originating from the steel substrate carry significantly reduced amplitudes compared to the acrylic surface echo.

Fig.4(c) shows the ultrasonic signal within the steel layer reflecting at the bottom interface with the water. Analysis using Equations (7) and (8) reveals that the steel-water interface acts as a soft boundary under these conditions. Consequently, the particle velocity of the reflected wave is in phase with that of the incident wave, while the acoustic pressure of the reflected wave is 180° out of phase with that of the incident wave. The negative sign of the acoustic pressure reflection coefficient (-0.93) in Table 2 confirms this phase reversal. At this interface, only a very small amount of acoustic signal enters the water layer, and almost all of the sound is reflected back into the steel.

Fig.4(d) illustrates the state after the ultrasonic signal in the steel has undergone one reflection at the bottom interface and subsequently another reflection at the steel–acrylic interface. Similarly, this interface also behaves as a soft boundary, causing the acoustic pressure of the reflected signal to undergo another 180° phase reversal. The remaining signal continues to transmit upwards to be received. It can be inferred that the primary and secondary echoes from the bottom surface of the steel each experience a 180° phase reversal at both soft boundaries. However, since both echoes undergo the same double reversal, their phases relative to each other in the time domain remain unchanged.

In summary, the acoustic impedance contrast between adjacent layers governs the energy distribution at each interface: a larger impedance difference leads to higher reflection and lower energy transmission. This progressive energy loss, combined with the differences in propagation path length, determines the temporal arrival sequence and amplitude decay of each echo. The two-dimensional distributions of acoustic pressure and acceleration in Fig.4 visually confirm this mechanism, and their consistency with the time-domain echo waveforms further validates the capability of this finite element simulation to accurately replicate ultrasonic wave propagation in multilayer structures.

IV. VERIFICATION OF ULTRASONIC ECHO SEQUENCE

A. Experimental environment

Figure.5 shows the experimental setup used to measure multiple echo signals from composite materials, which includes a DPR300 pulse generator, an oscilloscope with a sampling rate of 100 MHz, a water tank filled with degassed distilled water, and materials such as steel and acrylic. As illustrated in the

schematic of Fig.2, the acrylic block is placed on top of the steel block inside the water tank, forming the coating–metal composite structure. A sufficient water layer is maintained above the acrylic surface so that the ultrasonic probe remains fully immersed in water without direct contact with the test object. The ultrasonic probe is fixed using a level holder to ensure that it is perpendicular to the surface of the test object. After signal acquisition, the gain and output amplitude of the pulse generator should be carefully adjusted according to the observed waveform: excessively high power may cause signal saturation and waveform distortion, while insufficient power may result in weak echoes that make it difficult to distinguish multiple reflection signals. Then, using the echo mode of the pulse generator, multiple echo signals can be observed on the oscilloscope. The signals are collected and processed, resulting in the waveform shown in Figure 6(b).

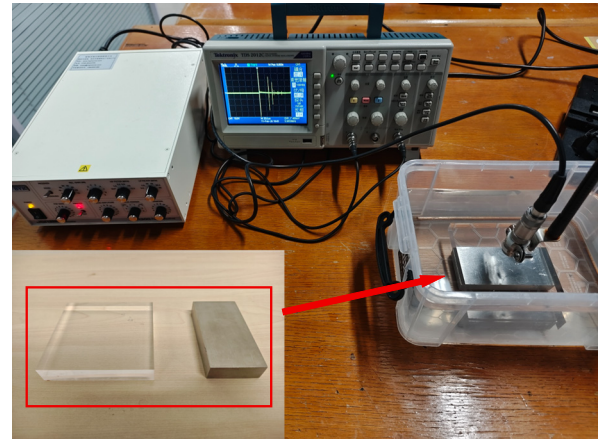


Fig. 5. Schematic diagram of equipment connection

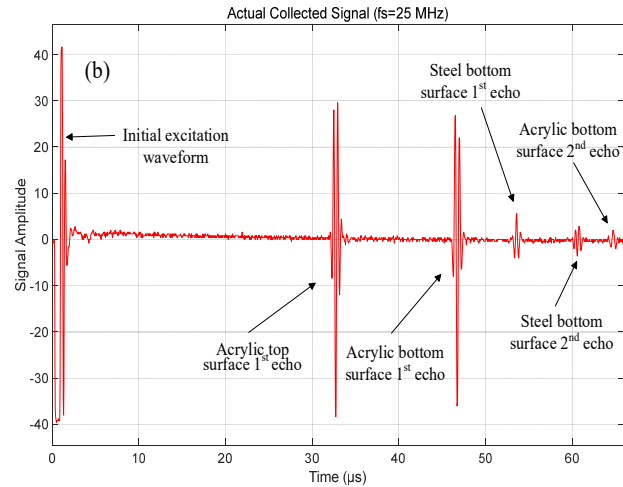
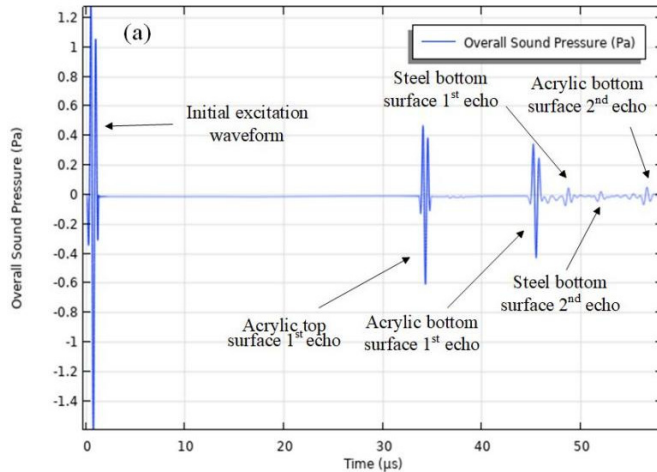


Fig. 6. Comparison of echo waveforms (excitation frequency: 2.25 MHz). (a) Simulated echo waveform (b) Actually collected echo waveform

B. Analysis of echo signal results

Using the excitation signal and initial parameters shown in Fig.3(b), the model with the external anti-corrosion coating is simulated. The simulation software calculates the corresponding time-domain waveform of the ultrasonic echo signals, as shown in Fig.6(a).

In Fig.6(a), the first signal is the pulse excitation signal located between $0.4 \mu\text{s}$ and $2.7 \mu\text{s}$, with the largest amplitude among all signals. The amplitudes of subsequent echoes are progressively reduced due to medium attenuation and interface transmission losses. The primary echo is the acrylic top surface first echo ($33.27\text{--}34.74 \mu\text{s}$), corresponding to the reflection at the water–acrylic interface. The second echo is the acrylic

bottom surface first echo (44.4–45.8 μs), which undergoes dual propagation attenuation through both the water and acrylic layers. Following these, the two smaller peaks appearing chronologically are the steel bottom surface first and second echoes.

Based on Fig.6(a), the time interval between the steel bottom surface first echo and the acrylic bottom surface first echo equals the interval between the steel bottom surface secondary echo and its first echo, both measuring 3.2 μs . This confirms that the secondary echo has traveled an additional round-trip distance equivalent to twice the steel thickness. Similarly, the time intervals between successive acrylic echoes are also equal, verifying the periodicity of multiple reflections within each layer.

By comparing Fig. 6(a) with Fig. 6(b), the order of receiving echo signals from different interfaces remains consistent between simulation and experiment. The slight differences between the simulated and experimental results are mainly attributed to several factors. First, the sound velocity and attenuation coefficients used in the simulation are derived from empirical measurements and material-property data, which may introduce uncertainty into the amplitude distribution and decay trend of the echoes. Second, although the probe is carefully fixed during the immersion test, its axis may not be perfectly perpendicular to the specimen surface, and even a slight angular deviation may affect the effective propagation path and lead to minor differences in echo amplitude, position, and waveform shape. In addition, the simulation adopts idealized excitation and boundary conditions, whereas the actual measurement may still be influenced by factors such as installation tolerance and environmental disturbance. Furthermore, due to computational resource limitations, the simulation cannot adopt the same time-domain length as the experimental signal, resulting in fewer observable echo repetitions in the simulated waveform.

Despite these limitations, the two-dimensional axisymmetric finite element model demonstrates clear advantages for analyzing ultrasonic propagation in multilayer media. By reducing the problem to two dimensions, the model significantly lowers computational cost while preserving essential wave propagation characteristics. Moreover, it provides dynamic visualization of acoustic pressure and acceleration fields at successive time instants, enabling direct observation of the spatiotemporal evolution of ultrasonic signals across multiple interfaces, which is difficult to achieve through experiments alone.

C. Frequency Domain Comparison of Multiple Echo Signals

To further validate the simulation beyond the time-domain comparison in Fig. 6, the acrylic top-surface first echo and the steel bottom-surface first echo were extracted from both the experimental and simulated datasets for frequency-domain analysis. A Hamming window was applied to both the experimental and simulated signals before FFT to suppress spectral leakage and improve the comparability of the spectral results.

Fig. 7 presents the time-domain waveforms and spectra of the simulated echoes. Both the acrylic top-surface first echo and the steel bottom-surface first echo show dominant spectral peaks near 2.25 MHz, indicating that the excitation frequency is

preserved during propagation. The steel bottom echo has a much lower spectral magnitude than the acrylic-surface echo, consistent with the cumulative attenuation and interface transmission loss in the multilayer structure.

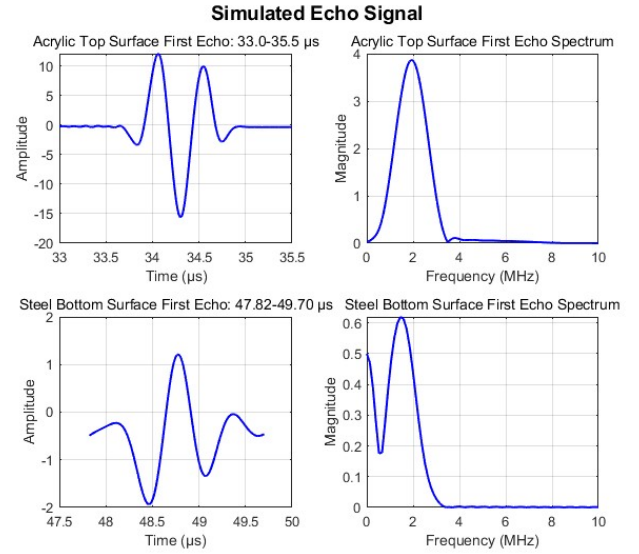


Fig. 7. Simulated echo signal truncation and spectrum

Fig. 8 shows the corresponding experimental results. The acrylic top-surface first echo also peaks near 2.25 MHz, while the steel bottom-surface first echo has a much smaller spectral magnitude, showing the same attenuation trend as in the simulation.

Comparing Fig.7 and Fig.8, both share consistent dominant frequency positions and similar amplitude decay patterns from the surface echo to the deeper echo. Minor differences in spectral bandwidth and secondary peaks can be attributed to the physical transducer bandwidth and material attenuation variations. This frequency-domain consistency confirms that the finite element model accurately reproduces the spectral characteristics of ultrasonic propagation in the multilayer structure.

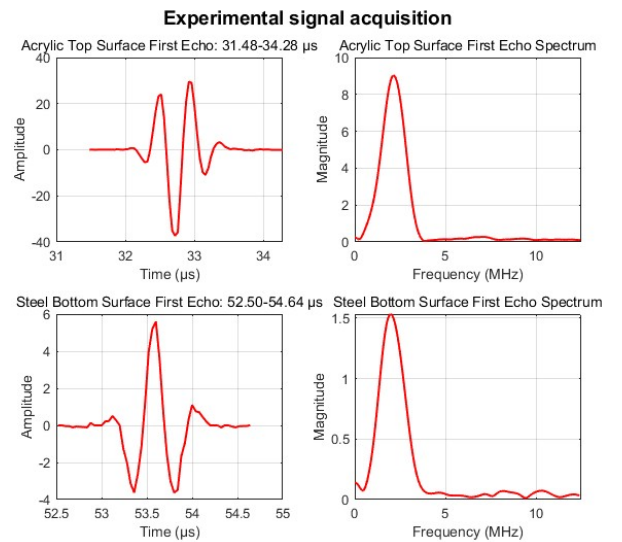


Fig. 8. Experimental echo signal truncation and spectrum

D. Simulated echo signals at different excitation frequencies

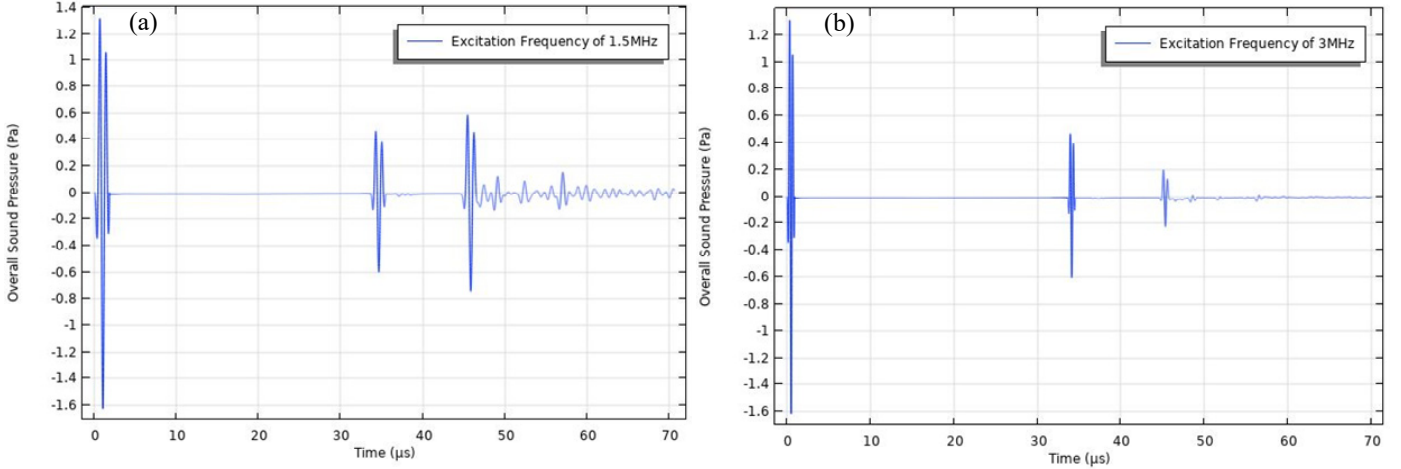


Fig. 9. Simulated echo signals at different excitation frequencies. (a) Excitation frequency 1.5MHz (b) Excitation frequency 3MHz

To investigate the influence of excitation frequency on ultrasonic propagation in the multilayer composite structure, simulations are performed at 1.5 MHz and 3 MHz. The excitation waveforms are derived by time-axis scaling of the original 2.25 MHz signal, and the resulting echo signals are presented in Fig.9.

As shown in Fig.9(a), the 1.5 MHz excitation produces broader echo waveforms, which reduces time-domain resolution. When the echo duration approaches or exceeds the inter-echo interval, overlap between successive reflections may occur, complicating the identification of individual layer boundaries. Conversely, as shown in Fig. 9(b), the 3 MHz excitation yields narrower waveforms with higher temporal resolution. However, the increased frequency also causes stronger propagation attenuation, resulting in lower echo energy and reduced signal-to-noise ratio, which makes signal detection less favorable.

Thus, excitation frequency selection involves a trade-off between temporal resolution and computational cost. For the composite structure investigated in this study, the 2.25 MHz excitation offers a balanced compromise, providing sufficient resolution to distinguish echoes from each interface while maintaining acceptable computational efficiency. These results offer a practical reference for selecting detection frequencies in engineering applications involving coated metal structures.

V. CONCLUSIONS

(1) A two-dimensional finite-element model was established to simulate the spatiotemporal propagation of ultrasonic waves in the submerged water–coating–metal multilayer structure. The model successfully reproduced the reflection, transmission, and multiple propagation processes at different interfaces, and provided the pulse-echo time-domain characteristics of the multilayer medium.

(2) By comparing the simulated results with the experimentally acquired signals in both the time domain and the frequency domain, the main echo sequence and dominant spectral characteristics were shown to be consistent. This agreement validates the feasibility of the simulation conditions

and confirms that the model can effectively describe ultrasonic propagation behavior in coated multilayer structures.

(3) The interface-coefficient analysis and multi-frequency simulation further clarified the influence of impedance mismatch and excitation frequency on echo amplitude, waveform resolution, and detectability. The results show that the proposed model is effective not only for echo mechanism interpretation, but also for providing practical guidance for parameter selection in ultrasonic inspection of coated metal structures.

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